

Antifungal Agents

Systemic & Topical

*Some are fungistatic,
while others are fungicidal*

Antifungal Agents

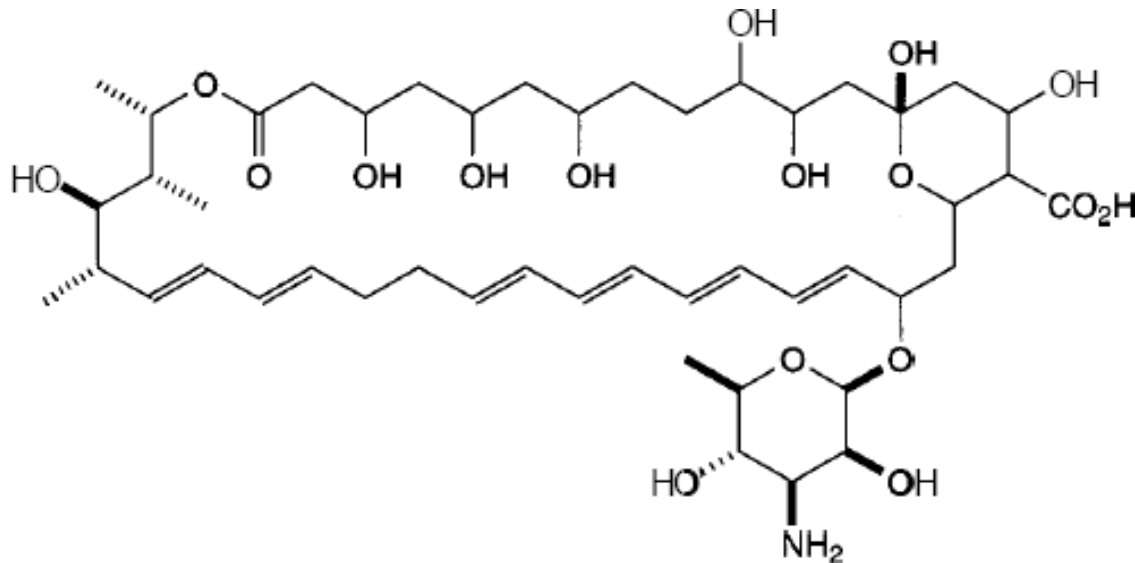
- An antifungal agent is a drug that selectively eliminates fungal pathogens from a host with minimal toxicity to the host.
- The development of antifungal agents has lagged behind that of antibacterial agents b/c
 - ✓ selectivity issue
 - ✓ fungi generally grow slowly and often in multicellular forms
- The following are the class of antifungal agents
 1. Polyene Antifungal Drugs
 2. Azoles Antifungal Drugs
 3. Allylamines Antifungal Drugs
 4. Other Antifungal Drugs

1. Polyene Antifungal Drugs

- The polyene compounds are so named because of the alternating conjugated double bonds that constitute a part of their macrolide ring structure.
- Polyenes display little or no activity against bacteria, as these organisms lack membrane-bound sterols.
- They are amphoteric, poorly soluble in water, and somewhat unstable because of extensive unsaturation.
- They include amphotericin B, nystatin, pimaricin, rimocidin and filipin
- MOA - These drugs interact with sterols in cell membranes (ergosterol in fungal cells; cholesterol in human cells) to form channels through the membrane, which leads to the leakage of intracellular components.
- They are fungicidal

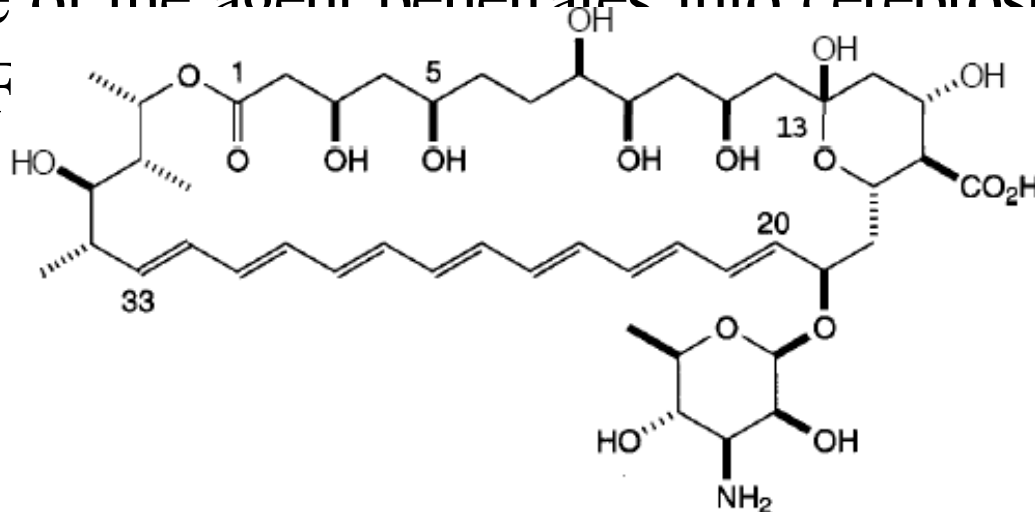
Nystatin

- Was the first successful antifungal antibiotic to be developed, and it is still in general use.
- The promise of its broad-spectrum antifungal activity is offset by host toxicity. Therefore, it is limited to topical use, where it has activity against yeasts such as the *Candida* species.



Amphotericin

- It has amphoteric properties and is insoluble in water at neutral pH
- Amphotericin B is the mainstay antifungal agent for treatment of life-threatening mycoses and for most other mycoses.
- Available in the form of intravenous injection and topical preparation.
- Little of the agent penetrates into cerebrospinal fluid (CSF)



(natamycin)

is active against both yeasts and moulds.



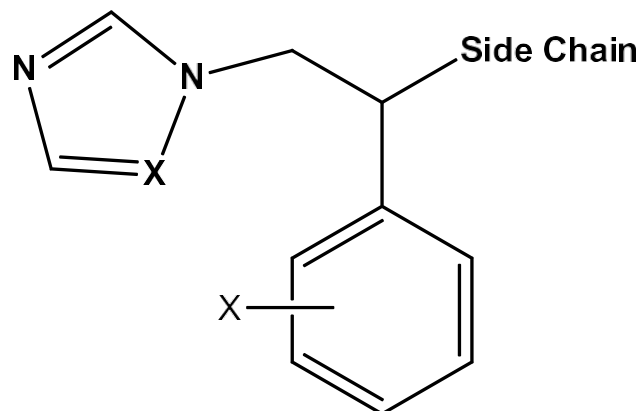
Structure Activity Relationships (SAR)

- The macrocyclic core with the conjugated double bonds, OH groups, a carboxyl groups and an aminosugar mycosamine are crucial for activity
- The hydrophilic, polyhydroxylated chain and the lipophilic polyene fragment are important
- Diversity among the polyenes arises principally from the number and configurational arrangement of hydroxyl groups and double bonds in the macrocycle.

2. Azoles Antifungal Drugs

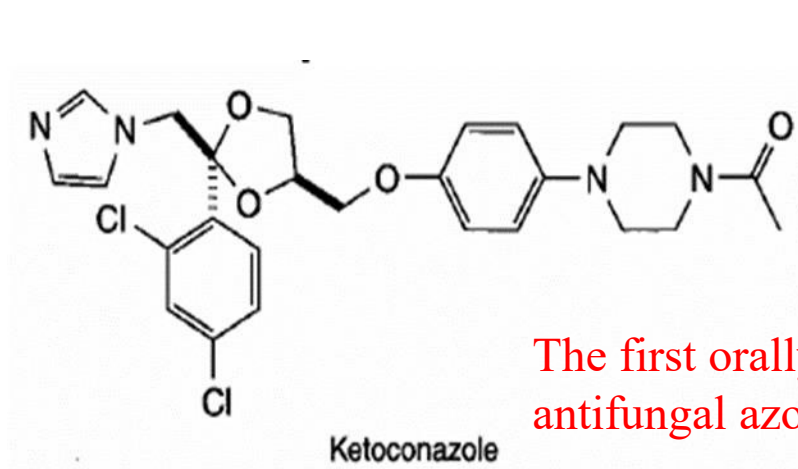
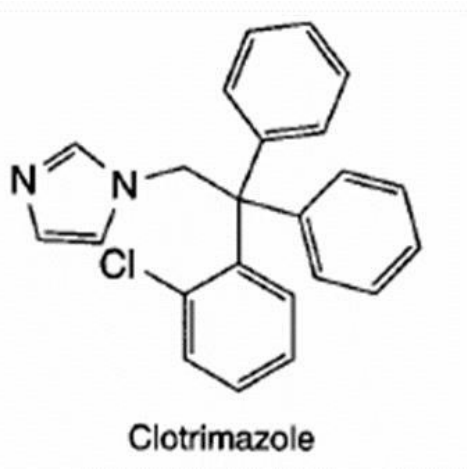
- Azoles have emerged as the preeminent class of antifungal agents
- The most significant agents in clinical use are chlormidazole, miconazole, clotrimazole, and econazole are administered topically; ketoconazole, itraconazole, and fluconazole are useful in the treatment of systemic infections.
- They work by inhibiting the biosynthesis of ergosterol
- The azole antifungal agents have five-membered organic rings that contain either two or three nitrogen molecules (the imidazole's and the triazoles respectively).
- These are fungistatic

Structure Activity Relationships

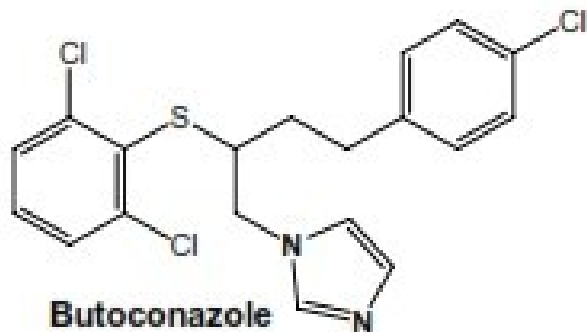
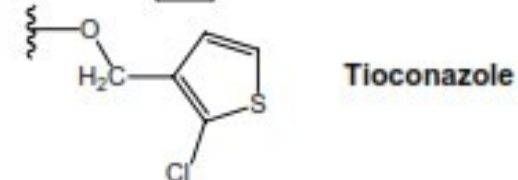
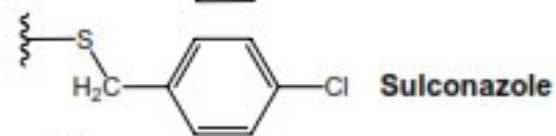
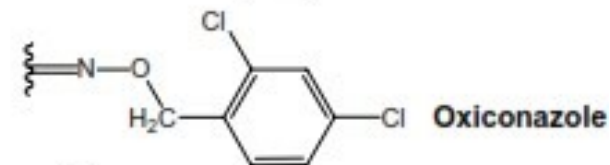
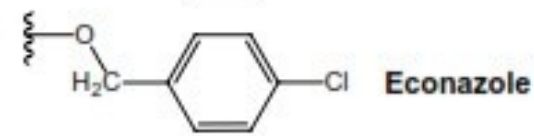
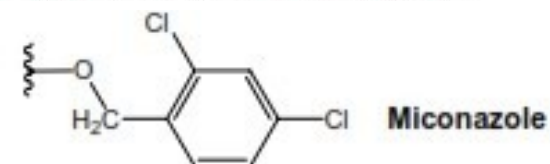
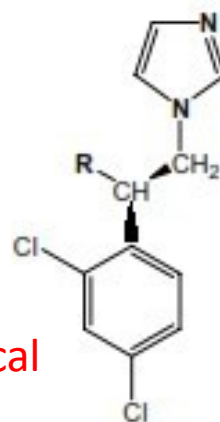
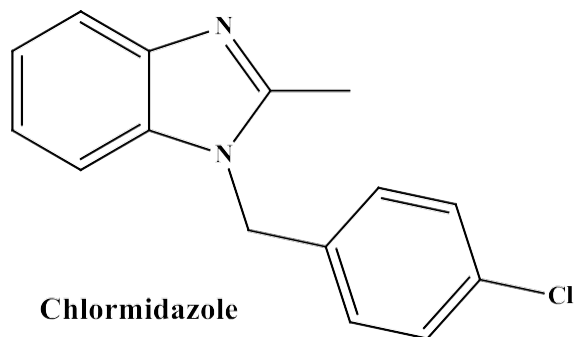


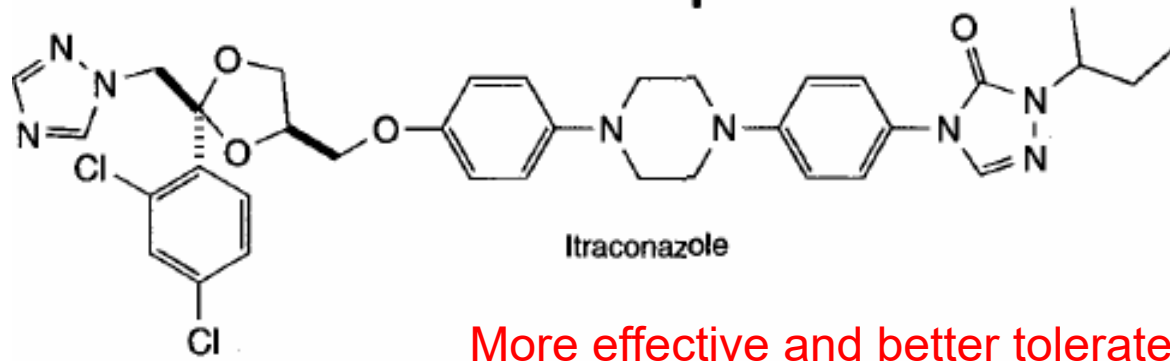
- A halo-substituted aromatic separated from the azole moiety by two atoms, and a side chain is important
- The most potent antifungal azoles possess two or three aromatic rings, at least one of which is halogen substituted (e.g., 2,4-dichlorophenyl, 4-chlorophenyl, 2,4-difluorophenyl), and other nonpolar functional groups.
- Only 2, and/or 2,4 substitution yields effective azole compounds.
- The halogen atom that yields the most potent compounds is fluorine, although functional groups such as sulfonic acids have been shown to do the same.

- Substitution at other positions of the ring yields inactive compounds.
- The side chain represents the feature of greatest diversity across the family
- The triazole or imidazole heme coordinating group is essential
- The activity $X=N > X=CH$
- The amidine nitrogen atom (N-3 in the imidazoles, N-4 in the triazoles) is believed to inhibit activation of molecular oxygen and prevent oxidation of steroidal substrates by the enzyme



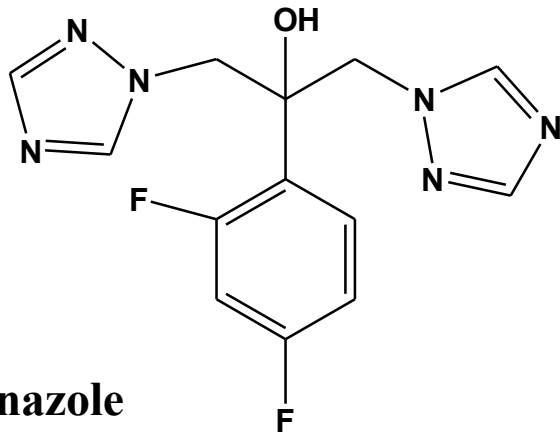
The first orally active antifungal azole





Itraconazole

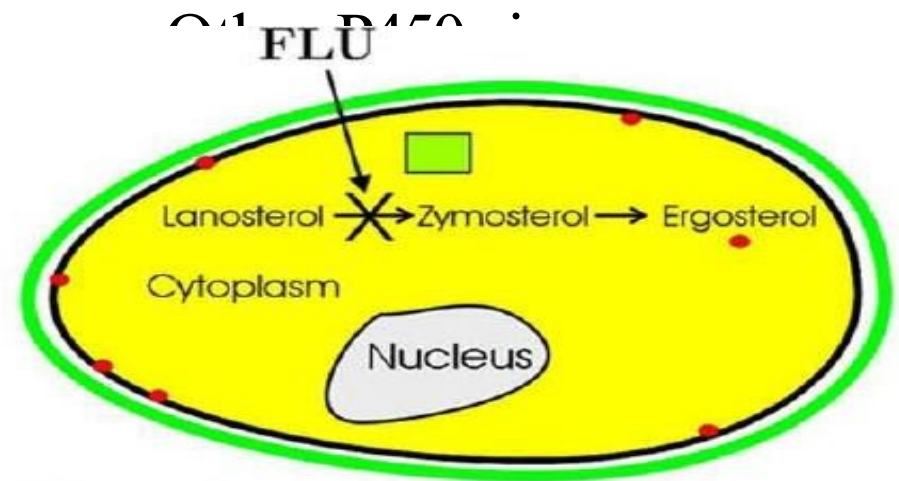
More effective and better tolerated, orally active, broad-spectrum antifungal agent than ketoconazole (but much more expensive)



Fluconazole

Two triazol rings + hydroxy group → water soluble → as a free base suitable for both oral and iv administration
Excellent oral bioavailability (90%) not affected by the presence of food or pH
Long half-life 27-34 h

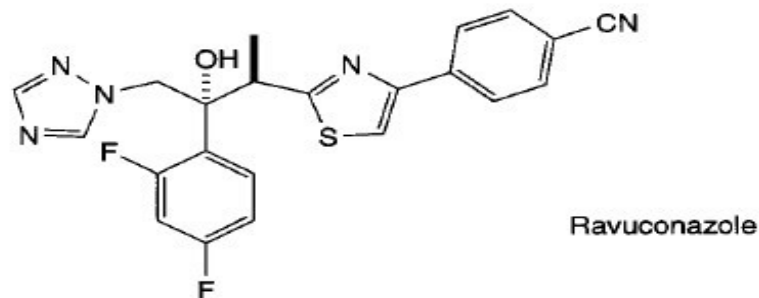
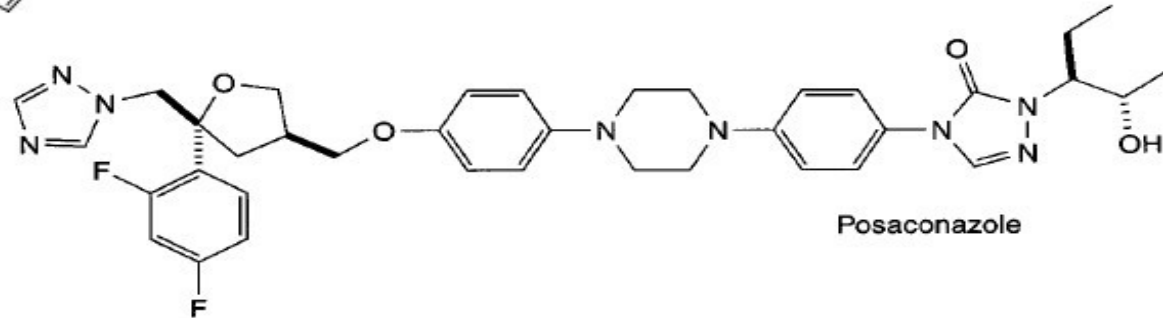
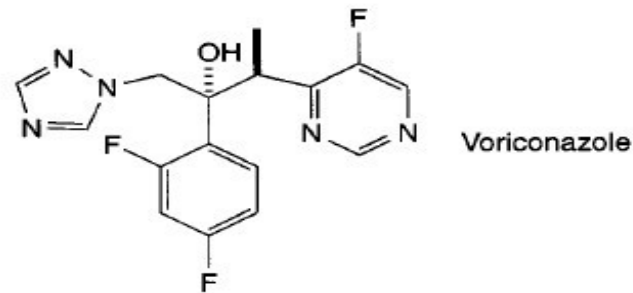
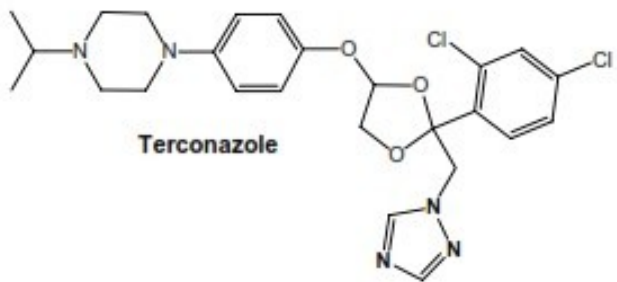
- Ketoconazole set the stage for the orally administered antifungal azoles. It can be administered both orally and topically
- The triazoles (fluconazole, itraconazole) have become the standard for the azoles, and have replaced amphotericin B for managing certain forms of the systemic mycoses
- Mechanism of Action: These imidazoles and triazoles inhibit CYP P450 14 α -demethylase in fungi. This enzyme is involved in the conversion of lanosterol to ergosterol.
- sterol biosynthesis may be affected



Newer

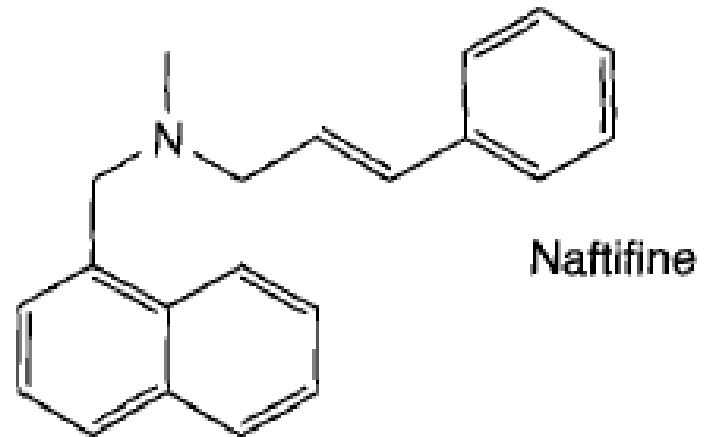
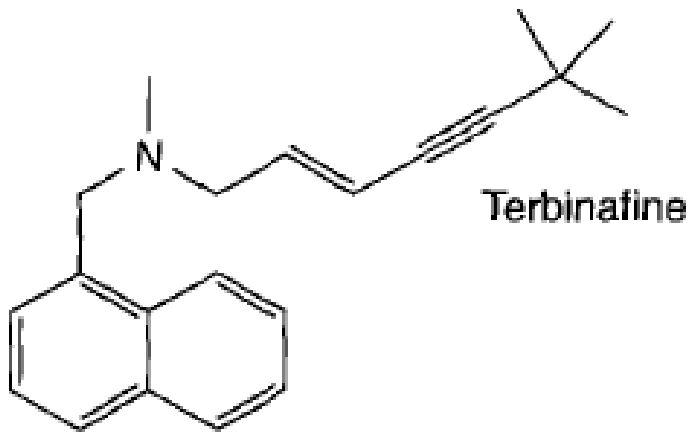
Antifungals

The three azoles that are currently under FDA review or in late-stage development are voriconazole, posaconazole, and ravuconazole.



3. Allylamines

- Allylamines (naftifine, terbinafine, butenafine, amorolfine) inhibit ergosterol biosynthesis at the level of squalene epoxidase.
- The intracellular accumulation of squalene that results is thought to be the primary cause of the fungicidal consequences of exposure to the drug



Structure Activity

Relationship

The structural requirements for potent activity are

- Two lipophilic domains linked to a central polar moiety by spacers of appropriate length;

For good activity,

- the polar moiety is a tertiary amine, and
- One of the lipophilic domains consists of a bicyclic aromatic ring system such as naphthalene or benzo[bithiophene].

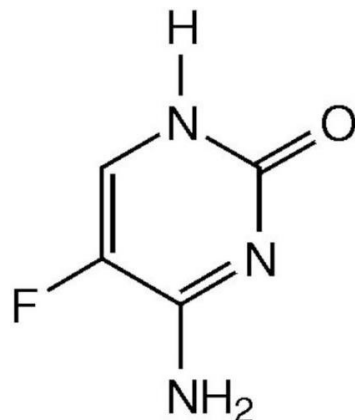
Mechanism of Action: These antifungal agents are reversible, noncompetitive inhibitors of the first step in ergosterol biosynthesis, the conversion of squalene to squalene-2,3epoxide by squalene epoxidase. The buildup of squalene in the cell membrane is toxic to the cell, causing pH imbalances and malfunction of membrane bound proteins.

4. Other common antifungal

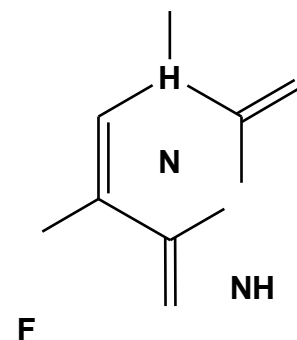
~~Fluor~~cytosine/Flucytosine

- Only available antimetabolite drug

Mechanism of Action: The drug enters the fungal cell through active transport on ATPases that normally transport pyrimidines. Once inside cells, fungal cytosine deaminase convert the drug to active 5- fluorouracil which is incorporated into RNA causing faulty RNA synthesis and also is a strong, non competitive inhibitor of thymidylate synthesis interrupting the one carbon pool substrate..



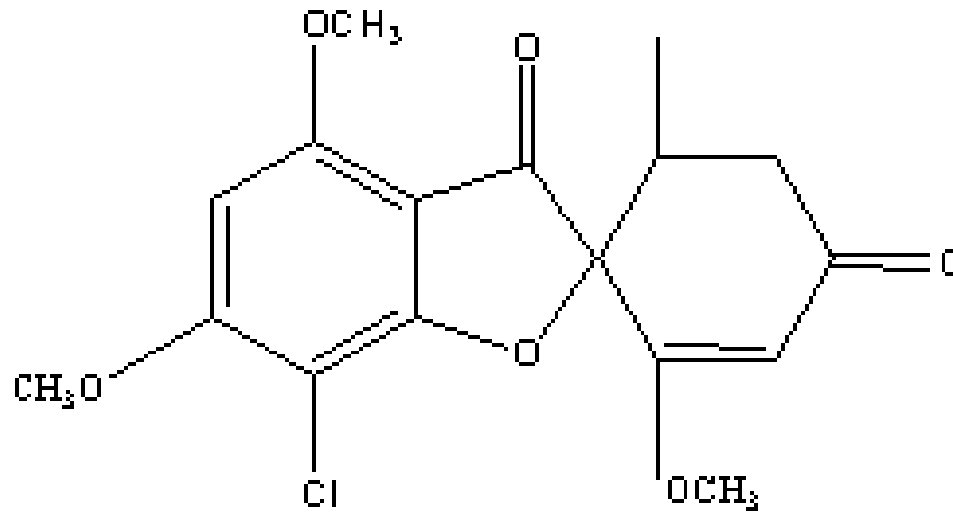
Fluorocytosine



Fluorouracil

Griseofulvin

- Inhibit fungal growth by binding to microtubules, disruption of mitotic spindles
- It is fungistatic



- *Mechanism of Action: Griseofulvin inhibits microtubule polymerization thus inhibiting the formation of the mitotic spindle.*

- **Ciclopirox** is a topical solution used to treat fungal infections of the nails and hair. It is a broad-spectrum antifungal medication that also has antibacterial and anti-inflammatory properties.

Mechanism of Action: Its main mode of action is thought to be its high affinity for trivalent cations, which inhibit essential co-factors in enzymes. Ciclopirox exhibits either fungistatic or fungicidal activity in vitro against a broad spectrum of fungal organisms.

